



September 2018 Examination

# EC400

## Probability and Statistical Inference

### Instructions to candidates

Time allowed: 2 hours

This paper contains **SEVEN** questions. Answer **ALL** questions.

On your script, write

- Your name
- Your class group number
- The name of your class teacher.

You are not allowed to use an electronic calculator

1. **(7 points - 10 minutes)** Let  $f(x) = (x + 1)/2$ ,  $-1 < x < 1$ , zero otherwise. Find

(a) The mode of the distribution.

(b) The 25th percentile (also called the first quartile) of the distribution.

2. **(8 points - 10 minutes)** If  $X$  has the following exponential distribution  $f(x) = \lambda e^{-\lambda x}$ ,  $0 < x < \infty$ , find the probability density function of  $Y$  where  $Y = \ln(\mu + \sigma X)$ .

3. **(12 points - 15 minutes)** Let  $X$  is a random variable with density

$$f(x) = \theta x^{\theta-1}, 0 < x < 1, 0 < \theta < \infty, \text{ zero elsewhere.}$$

(a) Obtain the expected value of  $X$ .

(b) A sample of  $n$  independent realizations of  $X$  ( $\{x_1, \dots, x_n\}$ ) was collected. Show that the method of moments estimator for  $\theta$  ( $\hat{\theta}_{MME}$ ) is  $\frac{\bar{X}}{1-\bar{X}}$ .

(c) Show that the method of moment estimator for  $\theta$  is consistent. Briefly argue that the estimator in general is biased.

4. **(13 points - 15 minutes)** Let  $(X, Y)$  have a joint density function

$$f_{X,Y}(x, y) = x + y, 0 \leq x \leq 1, 0 \leq y \leq 1, \text{ zero elsewhere.}$$

Derive the unconditional and conditional expectation of  $Y$  ( $E(Y)$  and  $E(Y|X = x)$ ) and discuss the relation between them.

5. **(12 points - 15 minutes)** The moment generating function (MGF), when it exists, completely determines the distribution of a random variable. Let  $M(t)$  be the moment generating function of the continuous random variable  $X$ .

(a) Provide a definition for the MGF  $M(t)$  that involves an integral.

(b) For all MGF's  $M(0) = 1$ . Let  $R(t) = \ln(M(t))$ . Show that the mean  $\mu = R'(0)$  and the variance  $\sigma^2 = R''(0)$ .

6. **(25 points - 30 minutes)** Let  $X$  equal the forced vital capacity (FVC) in liters for a female college student. (This is the amount of air that a student can force out of her lungs.). Assume that the distribution of  $X$  is (approximately)  $N(\mu, \sigma^2)$  and  $\sigma^2$  is unknown. A volleyball coach claims that the FVC of volleyball players is greater than 3.4 liters. She plans to test her claim using a random sample of size  $n = 9$ .

(a) Define  $H_0$  and  $H_1$ .

(b) Provide the test-statistic and give its distribution under the  $H_0$ .

(c) For the sample, she obtains  $\bar{x} = 3.7$  and  $s^2 = 1/9$ .

- i. Given a 5% level of significance, discuss whether you should reject the null or not. Interpret your finding.
- ii. Indicate how you would obtain the  $p$ -value of your test. Include a graph in your discussion.
- iii. Discuss how you would obtain the power of the test if the true value of volleyball players' FVC equals 3.8. You are not expected to provide an explicit numerical answer.

7. **(15 points - 18 minutes)** Let  $X_1, X_2, \dots, X_n$  be a random sample of size  $n$  from an exponential distribution with unknown mean equal to  $\theta$ . Using Moment Generating Functions it can be shown that the random variable  $W = \frac{2}{\theta} \sum_{i=1}^n X_i$  is  $\chi^2(2n)$ .

Let  $n = 7$  (clearly small)!

(a) Use  $W$  to construct a 90% confidence interval for  $\theta$ . You are told that 5th percentile and 95th percentile of  $\chi^2(14)$  are 6.56 and 23.68 respectively and  $\bar{x} = 93.6$ . (Suffices to provide the endpoint in terms of ratios that do not contain any unknowns).

*Comment:* Partial credit will be given to proving a clear definition of a 90% confidence interval for  $\theta$  only. Note: part (b) of the question, does not require the derivation of the confidence interval.

(b) Explain how this confidence interval may be used when considering a classical hypothesis test about the mean of  $\theta$ . Be sure you clearly indicate the  $H_0$ ,  $H_1$ , and the significance level you should adopt. Note, I am not asking for the test statistic itself.